Logometric Analysis of Russian Propaganda Narratives: A Quantitative Study of *ukraina.ru* (2014–2025)

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`github.com/Romain-Jaffuel/Russian-Propaganda-Analysis-Ukraina.ru`

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Abstract

Since 2014, the Russian Federation has deployed information warfare as a strategic extension of its military operations in Ukraine. Among the tools mobilized, the state-controlled website *ukraina.ru*—launched by the "Rossiya Segodnya" media group—serves as a long-term propaganda platform aimed at shaping perceptions of Ukraine's history, sovereignty, and internal politics. This paper presents a computational logometric analysis of over 11,000 *ukraina.ru* articles published between 2014 and 2025. Using natural language processing and unsupervised machine learning techniques (TF–IDF, TruncatedSVD and KMeans), the study models the semantic and temporal dynamics of Russian propaganda. Results reveal a synchronization between spikes in publication activity and the evolution of the conflict, along with a linguistic shift after February 2022. Terms like "denazification" and "Russophobia" appeared. Unsupervised topic clustering reveal how the articles emphasises on specific theme aligned with the Russian governement reading of Ukrainian history. These findings demonstrate how *ukraina.ru* is part of a broader startegy to undermine ukrainian unity in the actual war context.

Contents

1 Introduction

The Russian information strategy surrounding the war in Ukraine is a paradigmatic example of modern hybrid warfare, combining kinetic operations with psychological and informational influence. After the Euromaidan revolution in early 2014, Russia annexed Crimea and actively supported separatist movements in Donbas. Shortly after, on 18 June 2014, the Kremlin launched the online platform *ukraina.ru* under the supervision of Dmitry Kiselyov, director of the "Rossiya Segodnya" media conglomerate. Its official goal was to "objectively and promptly cover events in Ukraine," but the platform's discourse rapidly aligned with state propaganda narratives. (BBC News, 2014)

Information warfare, inherited from Soviet strategic culture, relies on control of narratives and manipulation of historical memory. Within this continuum, *ukraina.ru* became a central node of Russia's informational ecosystem, reproducing ideological frames such as the "Russian World" (*Русский Мир*) and "Near Abroad" (*Ближнее зарубежье*) (Pizzolo, 2020). These narratives merge national identity, imperial nostalgia, and moral justification for intervention.

1.1 Problem definition

The importance of media in propaganda was widely discussed. In their book, Jowett and O'Donnell (2019) stated: "Media utilization is vital to a propaganda campaign. Access to and control of the media literally means access to and potential control of public opinion." The Russian Federation is one of the most active actors in the field of propaganda, employing multidimensional and multilayered techniques.

Despite thousands of published articles, the discursive patterns of *ukraina.ru* remain largely understudied. Prior analyses of Russian disinformation often focused on social media or international outlets like *Sputnik* and *RT*, but not on domestic-oriented propaganda. Yet, *ukraina.ru* is explicitly targeted toward Russian and Russophone Ukrainian audiences (Arribas et al., 2023). The platform functions as both an ideological amplifier and a historical revisionism vector, merging current political framing with reinterpretations of Ukraine's past.

This study aims to model, classify, and interpret these mechanisms through computational analysis. The project introduces a large-scale dataset of more than 11,000 articles (7,000 opinion and 4,000 historical), providing a long-term quantitative mapping of Russian digital propaganda in the Ukrainian context.

1.2 Knowledge gap

While a growing body of research investigates Russian influence campaigns, few works quantify the historical narrative construction over time. As we know, no previous publication has analyzed *ukraina.ru*'s discourse evolution from 2014 to 2025, despite its consistent production. Our study fills this gap by proposing a quantitative linguistic analysis (frequency, topic modeling, clustering).

1.3 Research questions

This research addresses three main questions:

1. What historical and ideological notions are most emphasized across the articles?
2. How have these narratives changed since the full-scale invasion of February 2022?
3. How can *ukraina.ru* be situated within the broader framework of the Russian state's propaganda strategy?

1.4 Hypotheses

We hypothesize that the escalation of war in 2022 corresponds to a measurable linguistic shift in the *ukraina.ru* corpus, characterized by:

- Increased use of emotional and militarized vocabulary.
- A part of historical article focus on denying Ukrainian history and focus on soviet nostalgia.
- *ukraina.ru* serves as a semi-official canal to create and amplify anti-ukrainian narratives.

Through computational analysis, we expect to confirm that *ukraina.ru* functions is an adaptive instrument of Russian propaganda, evolving in coordination with military objectives and a multi-layered propaganda system.

2 Related Work

Nowaday, analysis of Russian propaganda has developed into a significant interdisciplinary field, combining media studies and computational linguistics. Other existing research highlights the diversity of propaganda techniques and their adaptation to geopolitical contexts.

2.1 Propaganda narratives and audiences

Polegkyi (2023) identifies four strategic propaganda narratives employed by the Kremlin, each addressing a distinct audience: domestic Russian citizens, Ukrainian Russophones, Western observers, and the Global South. These narratives intertwine victimhood, legitimacy, and civilizational mission to justify military actions. In this framework, *ukraina.ru* primarily targets internal Russians audiences and Ukrainians Russophone. It seeks to normalize the idea that Ukraine belongs to a shared "Russian world."

Similarly, Arribas et al. (2023) emphasizes that the linguistic and symbolic construction of disinformation in Russian media operates mostly in Russian, reinforcing the inward orientation of propaganda. Their study on history-based manipulation shows that such messages "seek to convince Russian citizens that Ukraine is an inseparable part of the country" (p. 16). This observation situates *ukraina.ru* as a cultural tool of nation-building through selective historical narratives.

2.2 Computational approaches to propaganda detection

From a computational standpoint, several works have developed machine learning approaches to identify propaganda techniques. Krak et al. (2024) and Krak et al. (2025) demonstrate that transformer-based architectures (RoBERTa) outperform classical methods such as logistic regression or recurrent neural networks in multi-label propaganda classification, reaching an average accuracy of 0.89 and up to 0.97 for certain techniques. Their methodology—combining the SemEval 2020 Task 11 dataset and fine-tuning for Russian—served as a reference for this study's classification phase.

In a complementary direction, S. Hanley and Durumeric (2022) reveal that 26.7% of *ukraina.ru* content originates from Telegram channels, highlighting the platform's integration into a broader disinformation ecosystem. Such findings suggest that the website acts as a hub, reprocessing narratives from decentralized sources into legitimized "journalistic" formats.

2.3 Propaganda and media ecosystems

Research by Sikorski (2023) on *Rubaltic.ru* demonstrates that regional propaganda outlets often mirror the Kremlin's central messages, amplifying them through localized frames. According to Springe (2017), even small pro-Russian portals (e.g., *Rubaltic.ru*, *Imho-club.lv*) manage to reach several hundred thousand readers monthly, with significant portions of their audience located in border states such as Lithuania and Latvia. This regional diffusion dynamic is consistent with *ukraina.ru*'s targeting of Russophone communities in Eastern and Southern Ukraine, with the singularity of a war context.

Field et al. (2018) and Paul and Matthews (2016) conceptualize Russian propaganda as a "firehose of falsehood": a model based on high volume, rapid repetition, and emotional resonance rather than logical coherence. This theory aligns closely with this study, which observes both informational saturation and thematic adaptability in *ukraina.ru*'s output.

2.4 Positioning of this study

Building upon these contributions, this research differs in two major aspects. First, it offers a diachronic quantitative perspective over more than a decade, allowing for temporal analysis of narrative shifts. Second, it integrates linguistic analysis (topic modeling, frequency analysis) on an unexplored Russian state-owned press website. This approach contributes to a broader understanding of the evolution of Russian propaganda applied to Russophone in a context of active war.

The next section presents the methodological framework adopted to collect, process, and analyze the data from *ukraina.ru*.

3 Methodology

This section presents the data collection, preprocessing, and pipeline applied to the *ukraina.ru* corpus. The workflow combines web scraping, data normalization, topic modeling, unsupervised clustering, and transformer-based classification to capture both semantic and temporal trends.

3.1 Data collection

All data were collected directly from the *ukraina.ru* website between May and June 2025. Because the site does not provide an API, scraping was conducted using `BeautifulSoup` and `Selenium` Python libraries. Each record contains the article's title, date, author, tags, hyperlink, and number of views.

A first pass retrieved the metadata of articles from the "History" and "Opinion" sections. However, due to inconsistent HTML structure before April 2020, a secondary script using Selenium was required to scrape 1,174 articles predating this format change. A subsequent script followed all links to extract full article content (title, text, publication date, tags, and view count) and stored it in JSON and CSV formats.

The resulting dataset includes 7,051 opinion and 4,087 historical articles, spanning from April 2014 to June 2025.

3.2 Data cleaning and preprocessing

All textual data were encoded in UTF-8 and processed using Python's `pandas`, `re`, and `nltk` libraries. The date field was standardized using the `datetime` module to enable temporal aggregation. Expressions such as "Вчера, 16:00" ("Yesterday, 16:00") or missing day values were resolved relative to the scraping date.

Normalization steps included lowercasing, removal of punctuation and HTML tags, and lemmatization with the Russian model of `pymorphy2`. Stopwords were removed using the `nltk.corpus.stopwords` list augmented with a custom list (e.g., "день," "который," "сегодня").

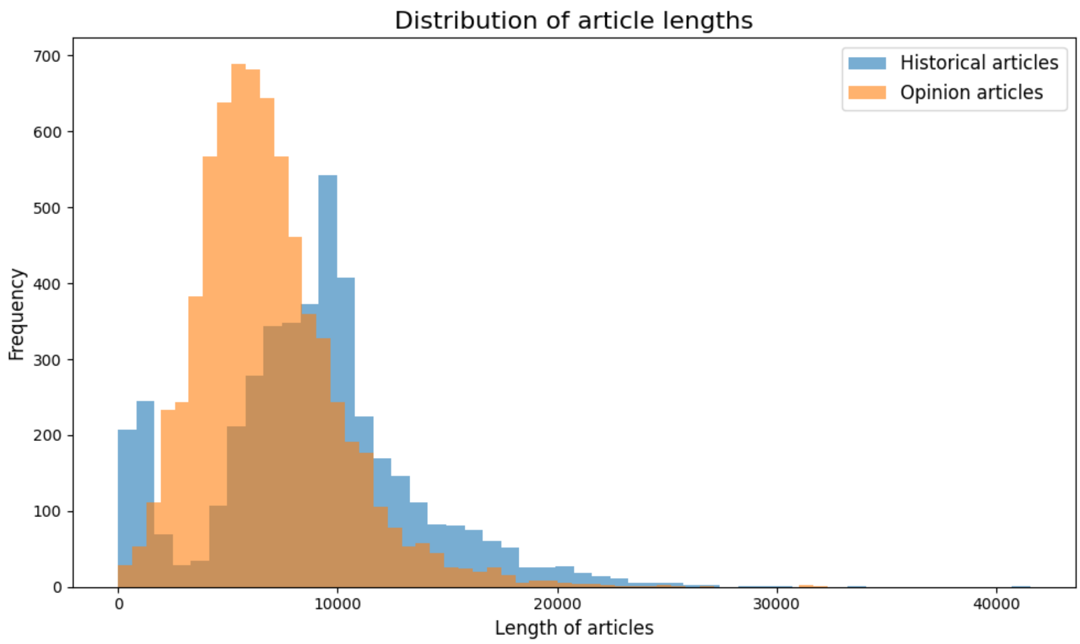

Figure 1: Distribution of article lengths (in characters) for historical and opinion sections.

As shown in **Figure 1**, the average Opinion article length is approximately 10,000 characters. The relative consistency of article size suggests a standardized editorial practice typical of state media production.

3.3 Temporal distribution of content categories

Temporal segmentation of the dataset reveals two distinct editorial rhythms. The opinion section maintained a continuous presence from 2014 to 2016 and then slowly increase again in 2017, while the historical section began in 2018.

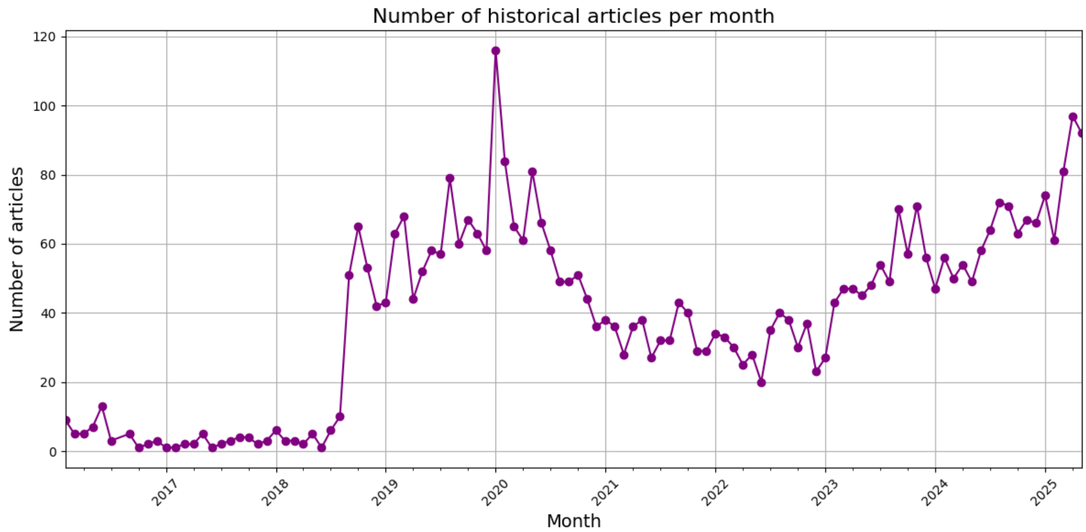

Figure 2: Number of historical articles per month (2016–2025).

Figure 2 highlights a surge in historical publications between 2019 and 2020, reaching a local maximum of 118 articles in November 2020. The number of articles has been increasing since mid-2022, correlating with the full-scale invasion.

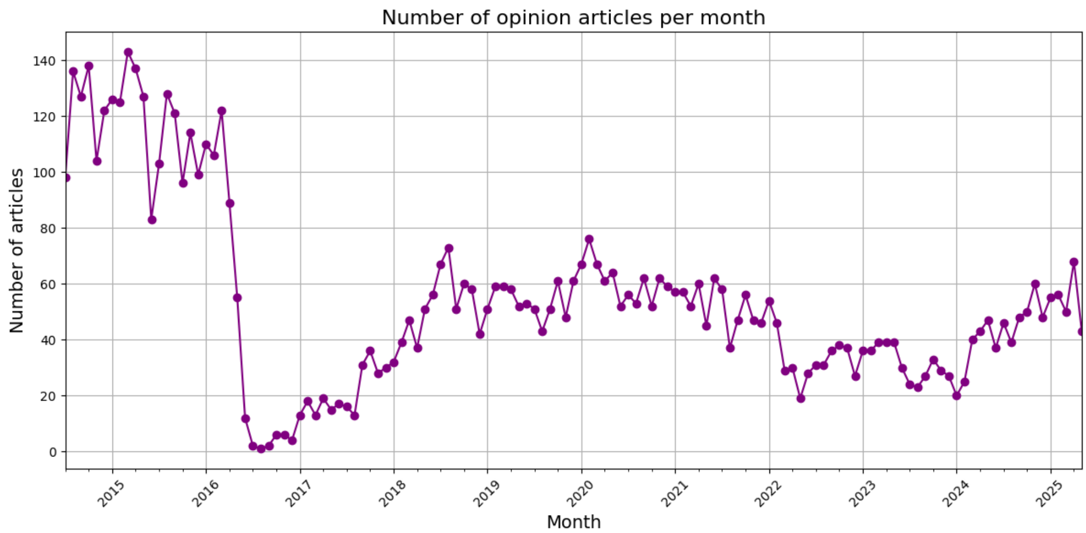

Figure 3: Number of opinion articles published per month (2014–2025).

Figure 3 presents a result about the number of opinion articles published each month. It reached its maximum in the number of publication during the first part of the conflict after 2014.

3.4 Topic modeling and clustering

Topic modeling was applied to identify major semantic domains across the historical corpus. After vectorizing texts using TF–IDF weighting, dimensionality reduction was performed with Truncated SVD (100 components) to enable clustering and visualization.

The TF–IDF representation was computed as follows (Salton & Buckley, 1988):

$$\text{TF} = \frac{\text{Number of times a word “X” appears in a document}}{\text{Total number of words in the document}}$$

$$\text{IDF} = \log\left(\frac{\text{Total number of documents in the corpus}}{\text{Number of documents where the word “X” appears}}\right)$$

$$\text{TF-IDF} = \text{TF} \times \text{IDF}$$

We then used `KMeans` algorithm and tested several numbers of clusters, with an evaluation based on the silhouette coefficient, defined as (Rousseeuw, 1987):

$$a(x) = \text{average distance of } x \text{ to all other points in the same cluster}$$

$$b(x) = \text{minimum average distance of } x \text{ to all points in other clusters}$$

$$s(x) = \frac{b(x) - a(x)}{\max\{\, a(x),\, b(x)\,\}}, \qquad s(x) \in [-1, 1]$$

$$SC = \frac{1}{N}\sum_{i=1}^{N} s(x_i)$$

We applied this evaluation method to every number of cluster from 2 to 30 :

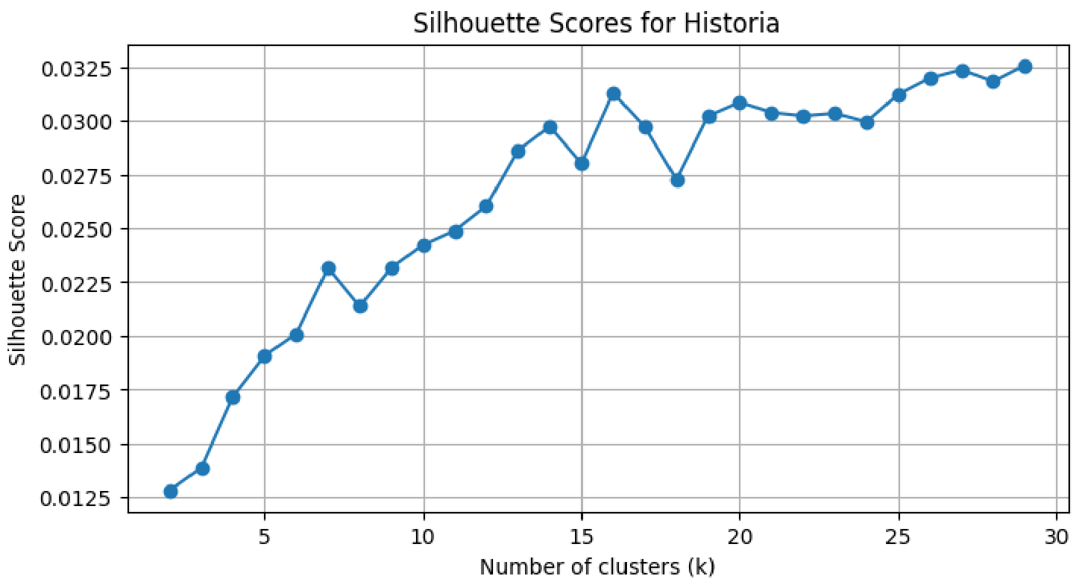

Figure 4: Silouhette Score for multiple number of cluster in Kmeans Method

Figure 4 shows the Silouhette Score of multiple number of Cluster tested with the

K-Means clustering method. We see a light spike at n=16 which is only reached again at 26 clusters, that is why we choosed it for our analysis. This number allows us to see the diversity of the topic discussed in the articles, without having too much similarities between clusters.

3.5 Supervised classification

To complement the unsupervised modeling, a zero-shot text classification approach was applied using the `facebook/bart-large-mnli` transformer model (Lewis et al., 2019). This model, trained on natural language inference (NLI) tasks, allows assigning a label to a text without additional fine-tuning. It estimates which hypothesis is most likely "entailed" by a given passage, making it suitable for ideological or narrative classification in the absence of manually annotated data.

Five labels were defined to capture the main interpretive frames found in the historical corpus: "Russian propaganda about Ukrainian history," "Historical revisionism," "Neutral account of history," "Fact-based historical narrative," and "Pro-Ukrainian historical narrative." For each text segment, the model was asked to evaluate the likelihood of the sentence "Этот текст — label." This Russian-language prompt reduced translation bias.

Because of input length limits, each article was divided into 1,000-character segments. Each segment was independently analyzed, and the model produced one probability score per label. The label with the highest probability was retained as the predicted category.

4 Results

This section presents the main results derived from unsupervised modeling and visual exploration of the *ukraina.ru* corpus. It synthesizes findings from both historical and opinion sections, focusing on temporal patterns, clustering outcomes, topic modeling, and the evolution of propaganda intensity.

4.1 Lexical patterns and semantic fields

The lexical distribution of the historical corpus shows a dense concentration of terms linking Russian and Soviet identity to Ukrainian history.

Figure 5: Most frequent terms across historical articles.

Figure 5 presents a frequency-based visualization of dominant terms. The most common words are “Россия” (Russia) and “СССР” (USSR). This lexical density empirically illustrates the triadic framing of Russia, Ukraine, and Soviet heritage as the backbone of historical discourse on the site.

4.2 Unsupervised clustering

Texts were vectorized using TF–IDF, reduced via TruncatedSVD to 100 dimensions, and clustered using optimal `KMeans` ($k = 16$).

Table 1: Main thematic clusters in historical articles identified by KMeans (2016–2025).

Cluster	**Dominant Topics**
C0	UNR, Petliura, Rada, 1917–1919, Bolsheviks, ZUNR.
C1	Cities, factories, monuments, housing, USSR projects.
C2	Poland, 1939, Western Belarus, minorities, territories.
C3	USSR, WWII, Germany, leadership, international politics.
C4	History, empire, Russian language, literature, identity.
C5	Jews, SS, 1941, Lvov, OUN, massacres, population losses.
C6	Eastern Front, Soviet Army, tanks, generals, campaigns.
C7	Churches, Kyiv, Moscow Patriarchate, Orthodoxy.
C8	Biographies, party cadres, veterans, revolutionaries.
C9	Kyivan Rus', princes, medieval statehood, expansion.
C10	Bulgakov, novels, Stalin era, writers, culture.
C11	Divisions, regiments, 1943 battles, commanders, fronts.
C12	OUN–UPA, Volhynia, NKVD, WWII ethnic violence.
C13	Cossacks, hetmans, Khmelnytsky, Commonwealth, Ottomans.
C14	Bessarabia, Romania, Russian/Ottoman wars, conflicts.
C15	Soviet cinema, films, actors, theater, war heroes.

Table 1 summarizes the sixteen dominant clusters identified within the historical corpus. Clusters C5 and C12 may suggest a negative association to the historical Ukrainian nationalist movement. And C4 may suggest a negation of Ukrainian cultural identity. Those components are part of the present-day military justification. However, further analysis are necessary to draw a conclusion.

4.3 Supervised classification

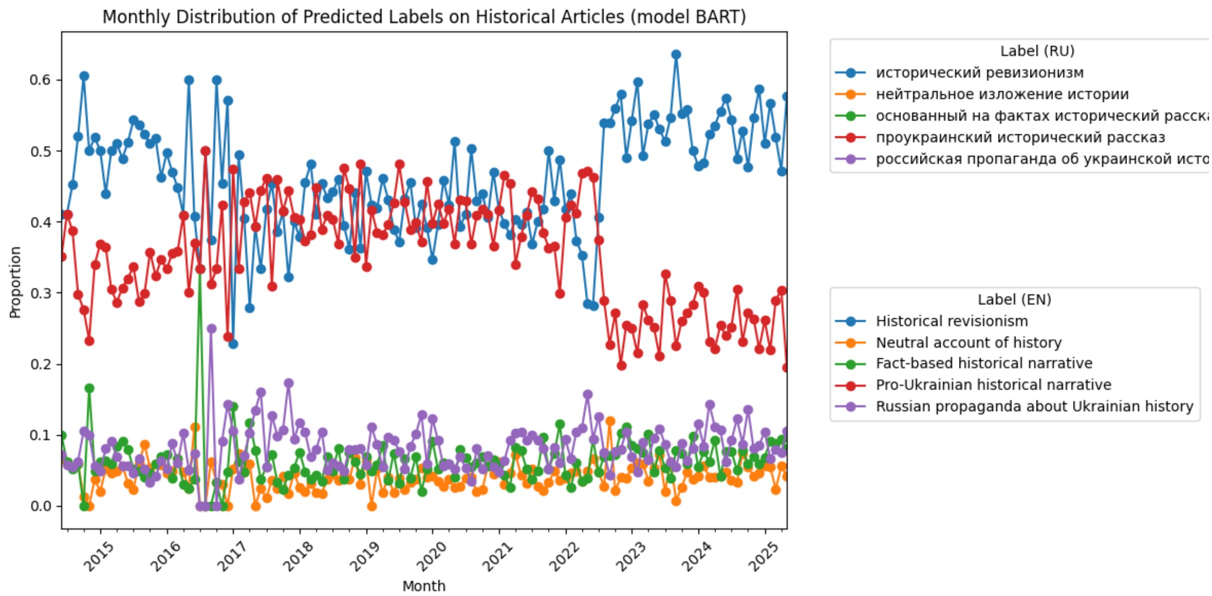

Figure 6: Monthly distribution of predicted propaganda labels (BART model).

Figure 6 presents the distribution of predicted labels by the BART classifier (Lewis et al., 2019). The figure shows a clear shift after February 2022 in accordance with the beginning of the war in Ukraine. The amount of historical revisionist articles was also high in 2014/2015, following the Donbas war and the annexation of Crimea.

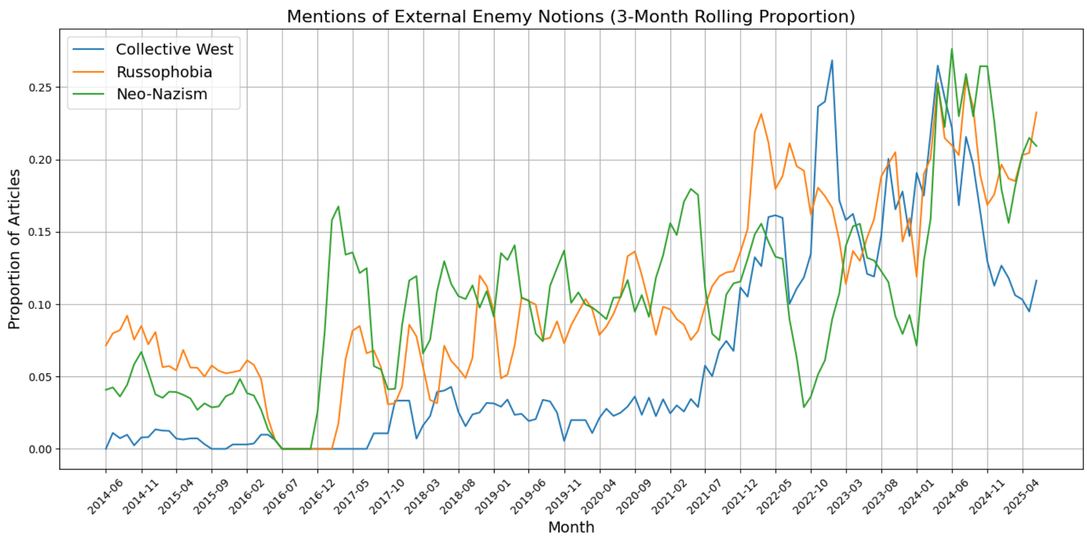

Figure 7: Mentions of "external enemy" notions (3-month rolling average).

Figure 7 shows the evolution of "external enemy" terminology such as "Collective West," "NATO," and "Russophobia" in the opinion articles. Those concepts were already used in the 2014–2015 period, but the proportion of use is every articles increased.

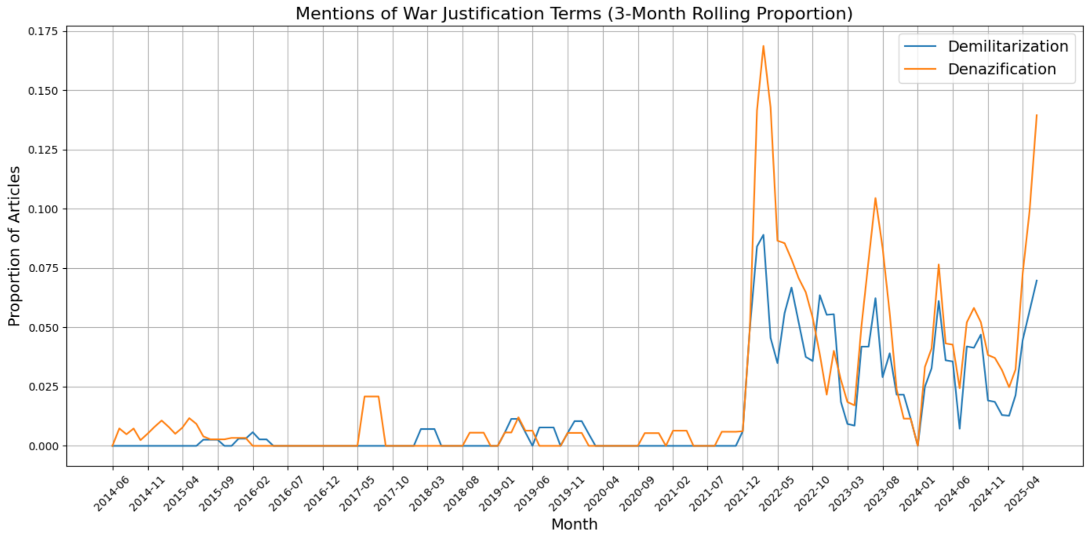

Figure 8: Mentions of war justification terms ("denazification," "demilitarization").

In **Figure 8**, war justification terms surged immediately after February 2022. This jump validates the hypothesis of linguistic synchronization with Kremlin narratives that appeared in the recent conflict but were not central for the 2014 conflict narrative.

5 Discussion

The computational findings highlight the adaptability of *ukraina.ru* as an instrument of information warfare. We will discuss the hypothesis that propaganda intensity functions as a strategic variable within hybrid conflict.

5.1 Narrative synchronization and multilayered propaganda

In their book, DeFleur and Ball-Rokeach (1982) largly discuss the importance of propaganda in a war context: "It became essential to mobilize sentiments and loyalties, to instill in citizens a hatred and fear of the enemy, to maintain their morale in the face of privation, and to capture their energies into an effective contribution to their nation.". Ukraina.ru is here to mobilize the sentiment of pro-Russia supporters throught the mobilisation of external threat and historical revisionism. The temporal analysis revealed coupling between editorial production and major political events. The shift in the narrative depicted in early 2022 mirrors what Field et al. (2018) describe as "agenda synchronization", the alignment of propaganda productions with real-world crises. And this can also be confirmed by the huge number of Telegram canal involved in the Russian propaganda system. From 1 October 2021 to 31 December 2023, 114,010 posts were made on the Ukraina.ru Telegram canal (NATO Strategic Communications Centre of Excellence, 2025). This leaves us with an average of 144 posts every days. A part of those post amplifies the propaganda made on a limited number of channels owned by Russian media outlets, including *ukraina.ru* press website. Moreover, H. W. A. Hanley and Durumeric (2024) revealed that 26.7% of *ukraina.ru* articles discussed content that originated or resulted from activity on Telegram. We can now conclude that the press website *ukraina.ru* work in pair with Telegram canals, and not only the official *ukraina.ru* Telegram canals. This link validates the "firehose of falsehood" model (Paul & Matthews, 2016), in which the effectiveness of disinformation derives not from credibility but from omnipresence and velocity. Telegram canals aim to occupy the digital space, while press website serve as a tool to strenghten a part of this quantitative propaganda.

5.2 Implications for computational propaganda analysis

By applying unsupervised clustering and transformer-based classification to the corpus, this study allow narratives analysis with quantitative reproducibility. The evidence supports a model of propaganda as an response to events included in defined structures.

These findings also help to understand how Russia frames its propaganda in a high intensity war context. Better understanding of the state-press propaganda in Ukraine can help to better counter a potential future application, for example against the Baltic states. Integrating such logometric indicators into real-time media observatories could enhance resilience against state-organized disinformation.

5.3 Conclusion

This study provided a comprehensive computational examination of *ukraina.ru*, a key component of the Russian state-controlled information ecosystem. By combining scraping, preprocessing, topic modeling, and transformer-based classification, it was possible to identify and quantify long-term narrative dynamics between 2014 and 2025.

The study also demonstrates how computational methods can operationalize concepts from information warfare theory, transforming qualitative notions into measurable linguistic features. Future research could extend this approach by incorporating cross-platform data (Telegram, VKontakte) and multimodal signals (images, videos) to evaluate coordination between multi-layered propaganda. Studies focusing on qualitative analysis of specific articles would also be relevant.

Ultimately, this project contributes to the emerging field of computational propaganda analysis by offering reproducible, data-driven evidence of how the Russian Federation weaponizes history, language, and information to sustain influence over domestic and foreign audiences.

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